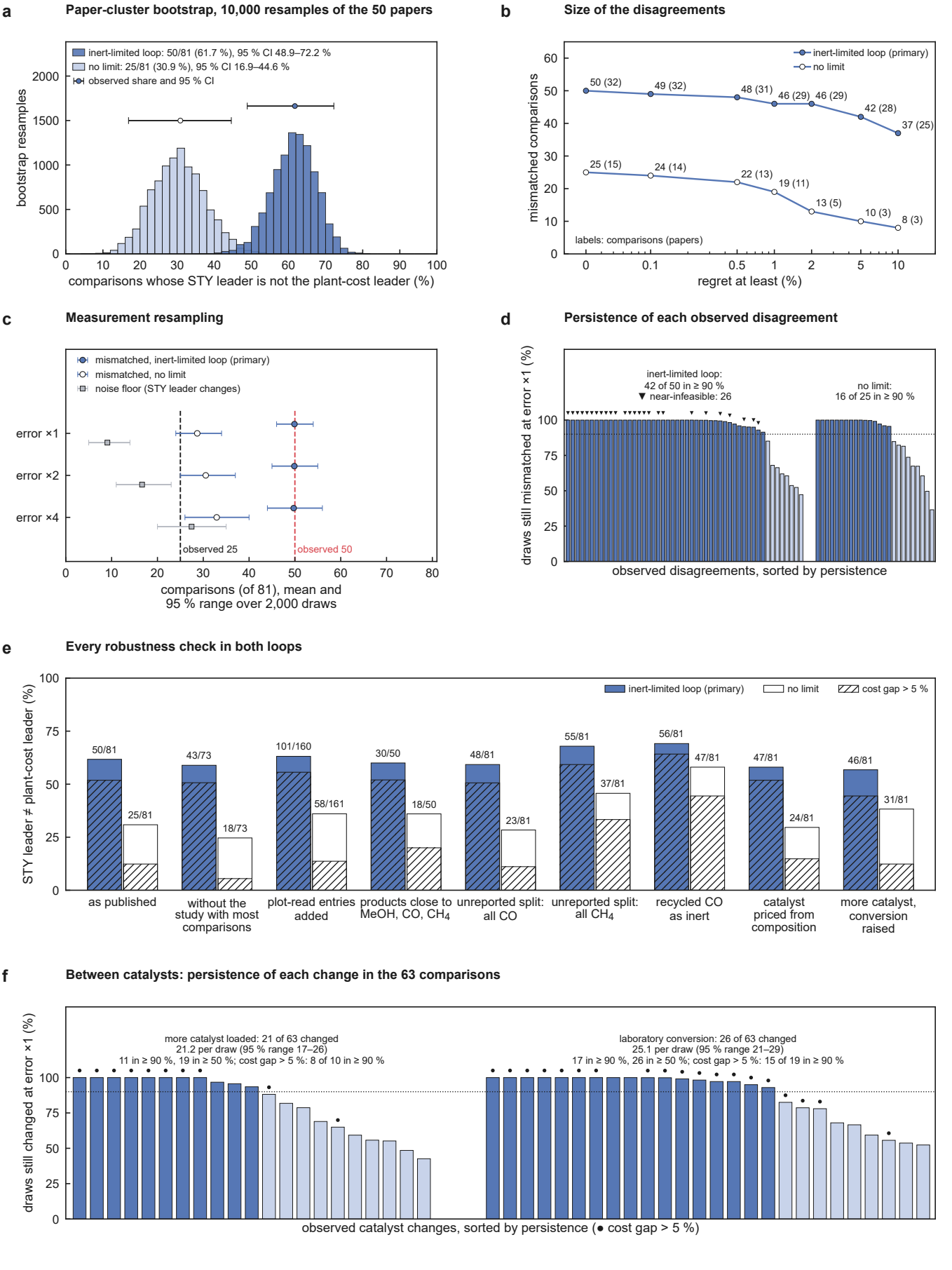

a
Paper-cluster bootstrap, 10,000 resamples of the 50 papers
inert-limited loop: 50/81 (61.7 %), 95 % CI 48.9–72.2 %
no limit: 25/81 (30.9 %), 95 % CI 16.9–44.6 %
observed share and 95 % CI
bootstrap resamples
comparisons whose STY leader is not the plant-cost leader (%)
b
Size of the disagreements
inert-limited loop (primary)
no limit
mismatched comparisons
50 (32)
49 (32)
48 (31)
46 (29)
46 (29)
42 (28)
37 (25)
25 (15)
24 (14)
22 (13)
19 (11)
13 (5)
10 (3)
8 (3)
labels: comparisons (papers)
regret at least (%)
c
Measurement resampling
mismatched, inert-limited loop (primary)
mismatched, no limit
noise floor (STY leader changes)
error ×1
error ×2
error ×4
observed 25
observed 50
comparisons (of 81), mean and
95 % range over 2,000 draws
d
Persistence of each observed disagreement
inert-limited loop:
42 of 50 in ≥ 90 %
▼ near-infeasible: 26
no limit:
16 of 25 in ≥ 90 %
draws still mismatched at error ×1 (%)
observed disagreements, sorted by persistence
e
Every robustness check in both loops
inert-limited loop (primary)
no limit
cost gap > 5 %
STY leader ≠ plant-cost leader (%)
50/81
25/81
43/73
18/73
101/160
58/161
30/50
18/50
48/81
23/81
55/81
37/81
56/81
47/81
47/81
24/81
46/81
31/81
as published
without the study with most comparisons
plot-read entries added
products close to MeOH, CO, CH4
unreported split: all CO
unreported split: all CH4
recycled CO as inert
catalyst priced from composition
more catalyst, conversion raised
f
Between catalysts: persistence of each change in the 63 comparisons
more catalyst loaded: 21 of 63 changed
21.2 per draw (95 % range 17–26)
11 in ≥ 90 %, 19 in ≥ 50 %; cost gap > 5 %: 8 of 10 in ≥ 90 %
laboratory conversion: 26 of 63 changed
25.1 per draw (95 % range 21–29)
17 in ≥ 90 %, 26 in ≥ 50 %; cost gap > 5 %: 15 of 19 in ≥ 90 %
draws still changed at error ×1 (%)
observed catalyst changes, sorted by persistence (● cost gap > 5 %)